Lecture 3: Multi-Omics data

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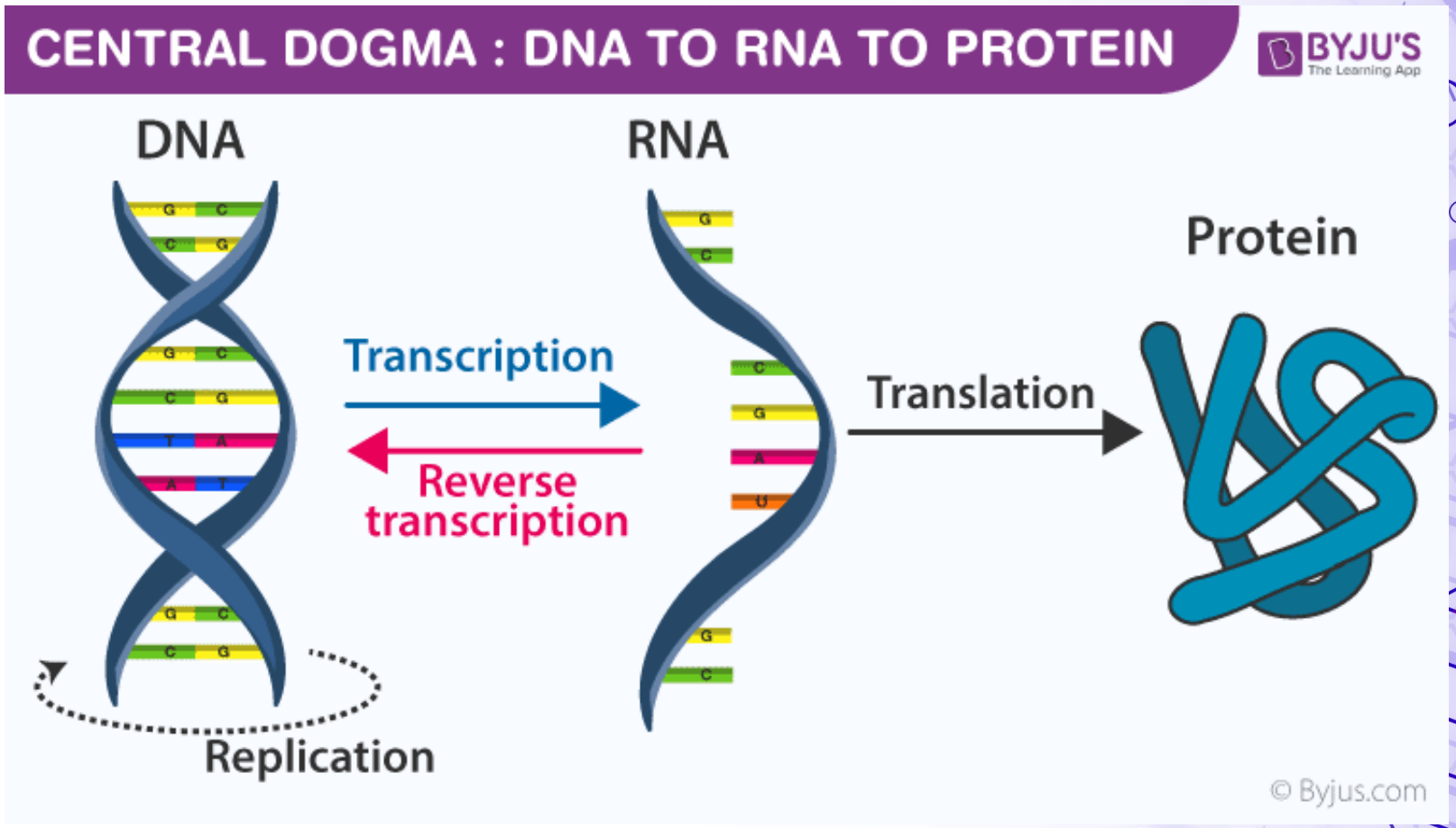
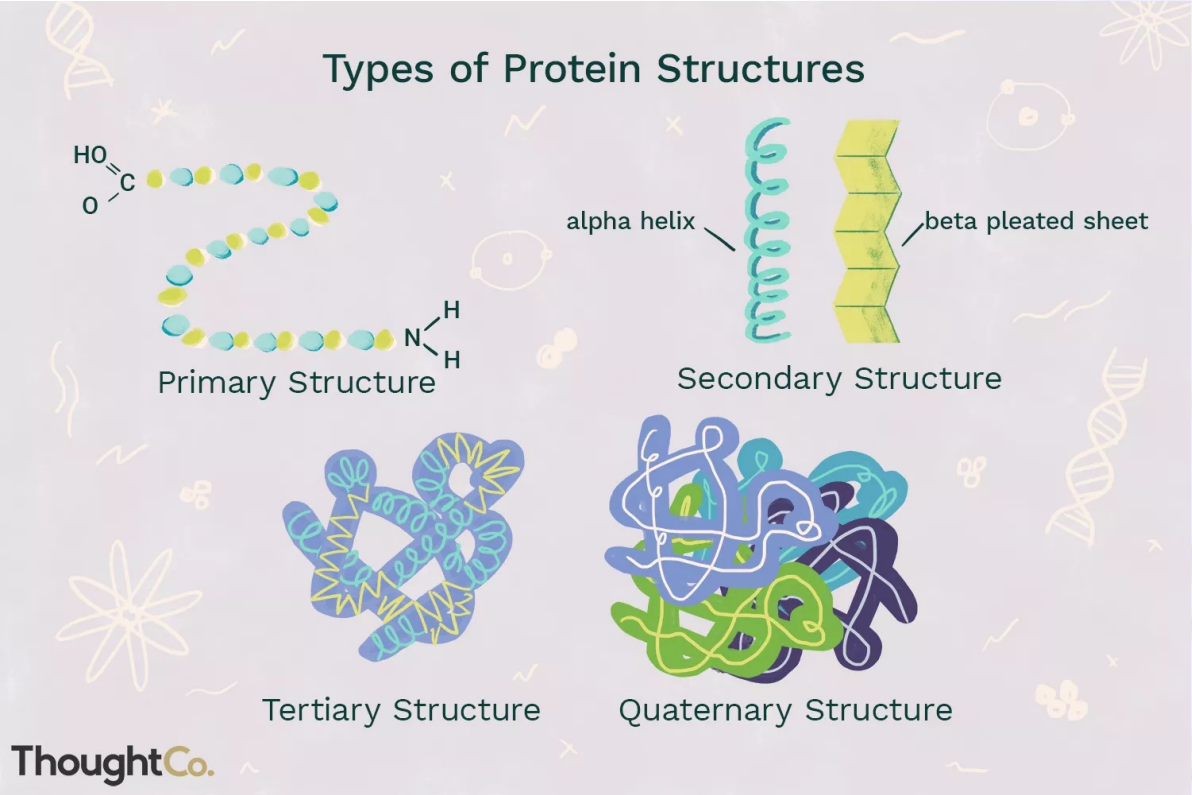


https://byjus.com/biology/central-dogma-inheritance-mechanism/

Protein structure: four levels



Primary structure (sequence): arrangement of 20 amino acids along the chain

Secondary structure: alpha helix, beta sheets

Tertiary: folding of the chain in 3D

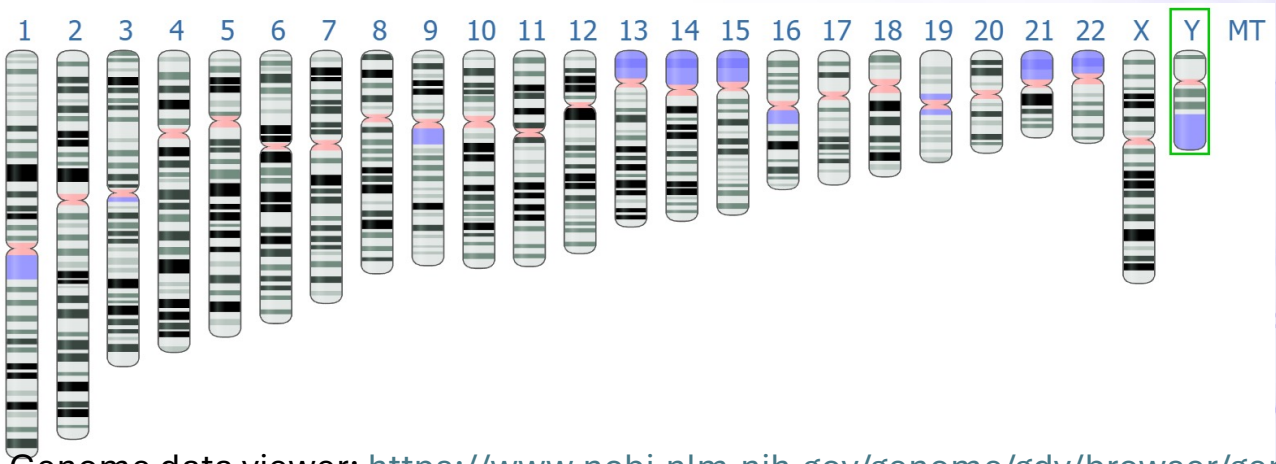
Quaternary structure: Stacking of chains

Genes , gene expression and gene regulations

- The genetic makeup of an organism is stored in its DNA, which is like an instruction book called the **genome.**
- **Genes** are specific parts of the DNA that give instructions to make proteins, like recipes in the book.
- Proteins do most of the work in the body — they build structures, carry messages, and keep everything running smoothly.
- But **not all genes are "on" all the time**. Cells turn genes on and off depending on what they need — this is called **gene expression**.
- Gene regulation is how the cell controls which genes are turned on or off, like using bookmarks and sticky notes to choose which recipes to use in the instruction book.

Percent of DNA translated to proteins

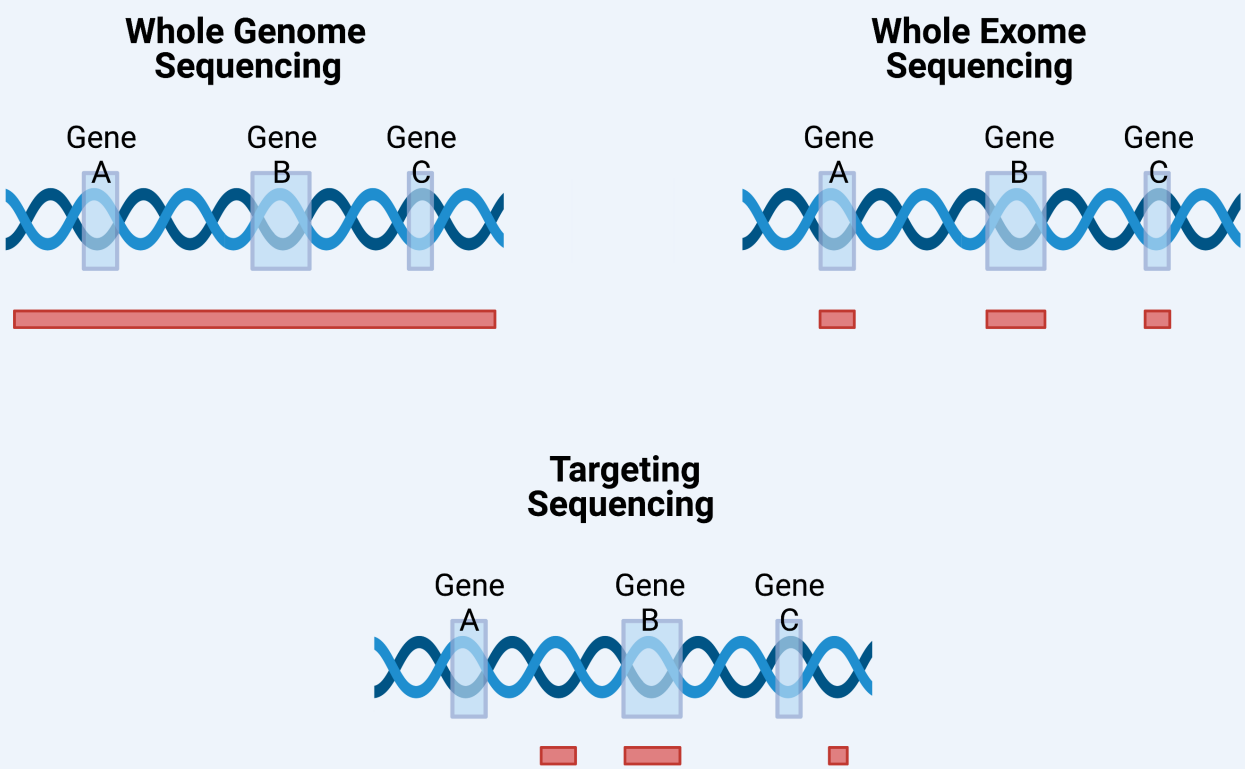
- The human genome contains approximately **3 billion base pairs**.
- Roughly **20,000 to 22,000 protein-coding genes** exist.
- These genes occupy about **1.5% to 2%** of the total genome.
- These coding regions are **transcribed** into mRNA and then **translated** into proteins.



Link to Genome data viewer: https://www.ncbi.nlm.nih.gov/genome/gdv/browser/genome/?id=GCF_000001405.40

Variations in Sequencing Approaches at Genome Levels

- Whole Genome Sequencing (WGS)
  - Sequencing whole genome, ~ 3 billion base pairs in humans, including coding and non-coding regions
- Whole Exome Sequencing (WES):
  - Sequencing DNA but only protein-coding parts (~2% of the genome, ~ 20,000 genes), focuses on mutations in genes that might affect protein functions
- Targeted Sequencing
  - **Targeted sequencing** relies on predetermining specific genes or DNA regions to sequence, often as part of a **gene panel**



https://www.fjc.gov/content/361254/dna-sequencing-and-what-it-can-reveal-about-dna-variation

Does your genome determine everything?

- Most Biological Traits (for example, how tall you will grow) are poly-genes trait, not a single gene effect.

- **Height is a complex trait controlled by hundreds of genes**, along with environmental factors like nutrition and health during childhood

- Your genome provides the blueprint, but it doesn't dictate the final building — your life, experiences, and environment all help shape what you become.

Personalized Medicine

- A medical approach that tailors treatment and prevention strategies to an individual’s genetic makeup.
- Moves away from "one-size-fits-all" treatments.
- Examples: predict risk factors for certain disease (e.g. BRCA1/2 mutations linked with breast cancers); Identifies which medications will work best for specific patients (pharmacogenomics).
- Build platforms that match drugs to patients’ genetic profiles.

Other Business Opportunities

- **Consumer Genetic Testing:** Provide direct-to-consumer (DTC) testing for ancestry, traits, nutrition, fitness, or skincare
  - Companies: 23andMe, AncestryDNA, MyHeritage.
- **Genomic Diagnostics:** Develop diagnostic tools for early detection of genetic disorders, cancers, rare diseases.
  - **Niche Markets**: Newborn genetic screening; Carrier screening for couples; Liquid biopsy companies for early cancer detection
- **Genetic Data Monetization: Example**: 23andMe has partnered with drug companies for research access.
- **Longevity & Preventive Health:** Use genomic risk prediction to tailor anti-aging or wellness programs.

Gene regulations at many levels

- Epigenetic level: involve DNA modification (DNA methylation), histone acetylation can alter the structure of chromatine, accessibility of genes
- Transcriptional regulation: This involves transcription factors that bind to promoters (regions of DNA sequence) or enhancers
- Post-transcriptional regulation: RNA processing, RNA stability, RNA transport
- Translational regulations
- Post-translational regulations (involving protein degradation)

Variation of sequencing technologies

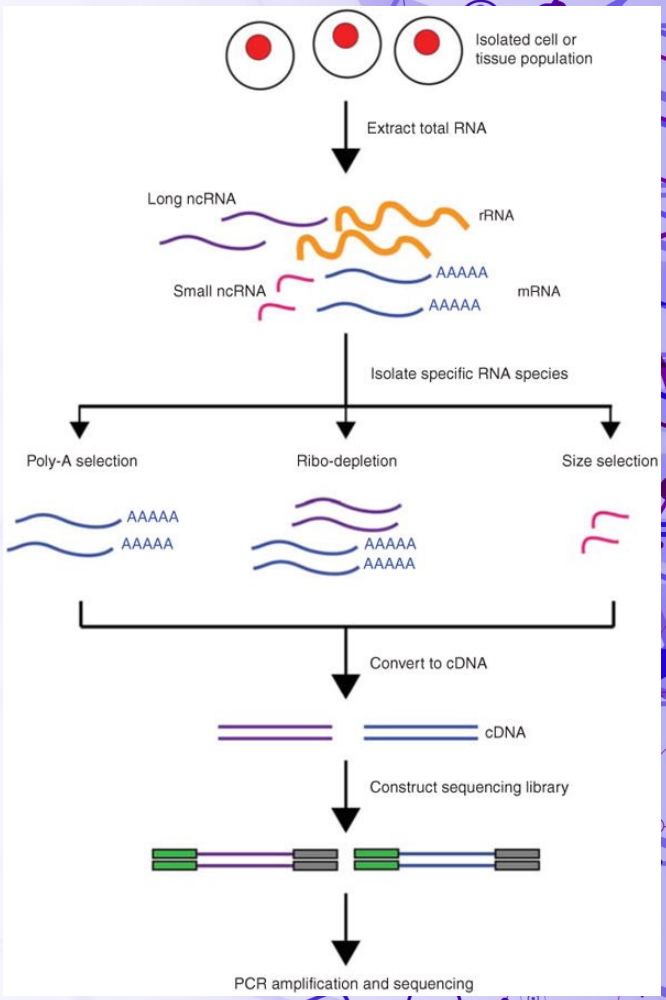
- RNA-sequencing (RNA-seq) :
  - Capture and quantify gene expression by sequencing messenger RNA (mRNA), identify differentially expressed genes, alternative splicing, and fusion genes
- ChIP-Sequencing (ChIP-Seq)
  - ChIP-seq stands for chromatin immunoprecipitation assays with sequencing. It pulls down DNA bound with transcription factors, histones. This allows for epigenetic regulation and transcriptional control
- ATAC-sequencing (ATAC-Seq)
  - ATAC stands for Assay for Transposase-Accessible Chromatin. IT is sequencing accessible chromatin in genome, revealing regions of open chromatin and potentially active in gene regulation
- Bisulfite Sequencing (WGBS or RRBS)
  - It detects presence of DNA methylation patterns in genome-wide, identifies epigenetic changes in tumorigenesis

How to measure gene expression?

- Can be measured at the protein-level
  - Western Blotting, ELISA, Proteomic Approach
- Can be measured at transcript-level (mRNA level)
  - Nothern blotting, Quantative PCR (qPCR), DNA microarray, RNA_seq
  - RNA-seq: RNA sequencing provides a comprehensive analysis of the transcriptome (all RNA molecules in a cell), allowing researchers to quantify gene expression levels and identify novel transcripts.

RNA-seq: A closer look

- RNA is extracted from the biological material of choice (e.g., cells, tissues).
- In most cell types, the majority of RNA molecules are rRNA, typically accounting for over 95% of the total cellular RNA
- Subsets of RNA molecules are isolated using a specific protocol, such as the poly-A selection protocol to enrich for polyadenylated transcripts or a ribo-depletion protocol to remove ribosomal RNAs.
- Next, the RNA is converted to complementary DNA (cDNA) by reverse transcription and sequencing adaptors are ligated to the ends of the cDNA fragments. Following amplification by PCR, the RNA-Seq library is ready for sequencing.



The Cancer Genome Atlas Project

The Cancer Genome Atlas (TCGA), a landmark cancer genomics program, molecularly characterized over 20,000 primary cancer and matched normal samples spanning 33 cancer types. This joint effort between NCI and the National Human Genome Research Institute began in 2006, bringing together researchers from diverse disciplines and multiple institutions.

- Over the next dozen years, TCGA generated over 2.5 petabytes of genomic, epigenomic, transcriptomic, and proteomic data. The data, which has already led to improvements in our ability to diagnose, treat, and prevent cancer, will remain publicly available for anyone in the research community to use.

The Cancer Genome Atlas(TCGA) Project

2005

Launch $100 million pilot

BRCA

2013

Pan cancer analysis group formed

2015

2016

Genomic Data Commons launched

2018

Pan-Cancer Atlas

Ref:
Cell-of-Origin Patterns Dominate the Molecular Classification of 10,000 Tumors from 33 Types of Cancer
Peter Laird and colleagues

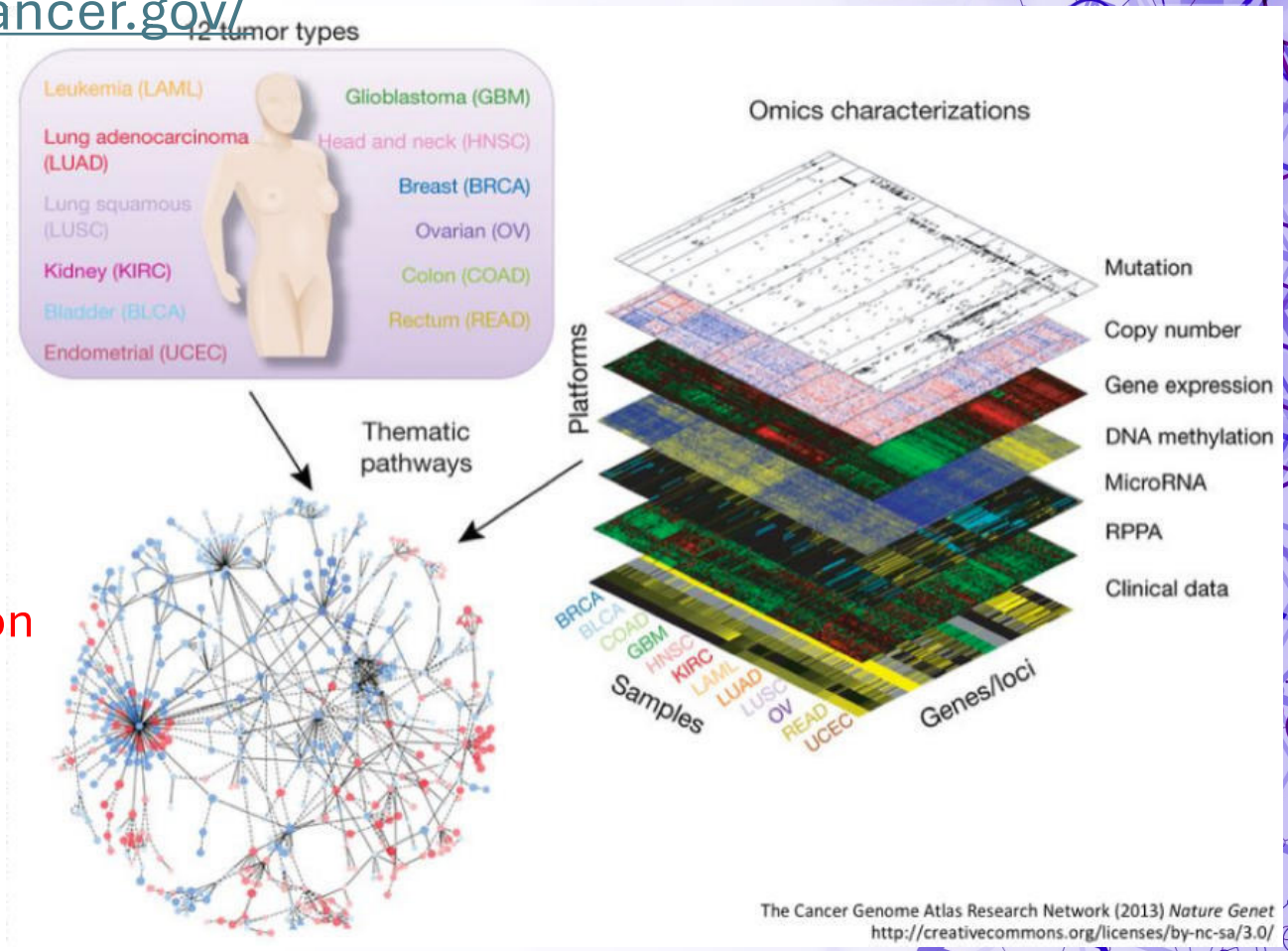
Where are these data stored

- Visit Xena Browser for TCGA processed data https://xenabrowser.net/datapages/
- Datasets are organized by Name of the Study
  - TCGA Breast Cancer (BRCA), TCGA prostate cancer (PRAD), TCGA Pancreatic cancer (PAAD)
  - This defines the types of tissues/patients/tumors
- Under each study, different datasets are available
  - Copy number of variations
  - DNA methylation
  - Gene expression by RNA-seq
  - Survival data (under phenotype data)
  - Protein expression by RPPA

Genomic Data Commons (GDC: 2016)

https://portal.gdc.cancer.gov/

Mutations
DNA methylation
Gene expression
Copy Number Variation



The Cancer Genome Atlas Research Network (2013) *Nature Genet*
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Thinking of Data in terms of Table

- All data are stored in Tables
- Important things to pay attention: Table Row Labels, Table Column Labels, Table cell values

Tissue	Gene expression levels					Diagnosis
	Gene 1	Gene 2	Gene 3	Gene 4	Gene 5	
1	0.405	0.326	0.234	0.348	0.748	normal
2	0.089	0.293	0.192	0.123	0.385	normal
3	0.459	0.125	0.543	0.334	0.218	tumor
4	0.123	0.389	0.238	0.651	0.972	normal
5	0.951	0.040	0.490	0.283	0.321	normal
6	0.297	0.859	0.219	0.783	0.984	tumor

Gene expression by RNA-seq

- Captured mRNA sequenced and aligned to the genome
- The expression of a gene is measured by the number of mRNA transcripts captured/aligned
- Gene names are given by ID/Gene mapping probe name

You will be trained to use Python code to read data and manipulate data in Jupyter Notebook

Be able to explore gene expression/gene mutations in cancer

Recommended Courses—advanced training

- Applied Data Science with Python Specialization [5 courses] (UMich) | Coursera
- Genomic Data Science Specialization [6 courses] (Johns Hopkins) | Coursera

Blog to prepare for this lecture

- Do a google search on some companies that build their business using sequencing technologies and reflect on their business model (List of potential companies: 23&me, **Tempus, Helix, GRAIL). Write a blog talk about the company and be sure to include**
  - Who is the founder of the company
  - When the company was founded? How much initial investment? What is the current revenue?
  - What is their main business focus?
  - What is the future success of the company?

Python Coding Activity

- Download an excel file that contains gene expression of a group genes called integrins (Gtex_integrin_7_organs.xlsx); read the data using panda.read
- Integrins is a set gene responsible for cell-cell communications. Do a little search to learn what is integrin.
- Filter the data by organ; for example you just want the Brain
- Make violin plots of all the genes for (a) Brain; (b) Liver ; Make a split violin plot
- Make a tSNE plot using all the data, label by their organ sites.